

Test Cases and Validation Report

Week 6 Final Capstone | Petalnex Pvt. Ltd. AI Automation Internship

Testing Strategy

- Normal cases verify the happy path.
- Edge cases verify missing, duplicated and oversized data.
- AI failure cases verify structured output validation and fallback.
- Integration failures verify database and email error handling.
- Security cases verify secret protection, prompt injection resistance and fair evaluation.

Acceptance Rule

A test passes only when the actual workflow behavior matches the expected outcome and does not create an unsafe or ambiguous final candidate state.

Test Matrix

ID	Scenario	Input / Action	Expected Result	Result
TC01	Valid strong candidate	Complete valid application with required skills	Candidate stored, AI runs, review required	Pass
TC02	Valid weak candidate	Complete application lacking several required skills	Candidate stored and sent for human review	Pass
TC03	Missing name	Email and skills supplied, name absent	Validation rejects request; no AI call	Pass
TC04	Missing email	Name and skills supplied, email absent	Validation rejects request; no email sent	Pass
TC05	Invalid email	candidate_email = abc	Validation rejects request	Pass
TC06	Empty skills	Skills field blank	Validation rejects or flags incomplete application	Pass
TC07	Very long input	Oversized CV/profile text	Input limit or controlled truncation prevents abuse	Pass
TC08	Duplicate application	Same email submitted twice	Duplicate is detected or safely recorded without accidental overwrite	Pass
TC09	AI valid output	Gemini returns schema compliant JSON	Structured output accepted	Pass
TC10	AI malformed output	Gemini returns invalid JSON/value	Parser/fallback blocks invalid output	Pass
TC11	AI recommendation review	AI returns review	Human review remains mandatory	Pass
TC12	Human shortlist	Reviewer selects shortlist	Final status = shortlisted; interview/email path runs	Pass

ID	Scenario	Input / Action	Expected Result	Result
TC13	Human reject	Reviewer selects reject	Final status = rejected; rejection email path runs	Pass
TC14	AI and human disagree	AI says reject, human says shortlist	Human decision wins; audit preserves both values	Pass
TC15	Human invalid value	Reviewer enters pending/other invalid value	Normalization blocks it until shortlist or reject	Pass
TC16	Database failure	Supabase insert/update unavailable	Error path logs failure and alerts administrator	Pass
TC17	Gmail failure	Email node fails	Database records email failure; admin path is triggered	Pass
TC18	Credential exposure	Inspect execution data and exported workflow	No API keys/passwords/secrets appear	Pass
TC19	Prompt injection in CV	CV contains instructions attempting to override evaluator	AI treats CV as data, not workflow instructions	Pass
TC20	Protected attribute bias	CV contains demographic/protected information	Evaluation remains job-relevant and does not use protected traits	Pass